

How to Improve the Performance of Drupal Sites

A site is only as fast as its last mile connectivity. If users access a site from a slow connection, even a site capable of responding quickly to requests will appear to be slow. A content management system like Drupal throws in additional challenges to the Web architect in improving performance because, typically, the Apache process takes up more space than a site serving traditional HTML or PHP pages. Read on to take a look at the various external factors that impact performance, and explore ways to mitigate them.



If customers complain of lengthy page load times, browsers displaying white pages, and 408 Request Timeout errors, then these are tell tale signs indicating that your site infrastructure is unable to cope up with the traffic, usually resulting in the server taking a long time to respond to a request or sometimes not returning the requested pages.

Drupal site performance bottlenecks: Where does the time go?

The first step is to identify the bottlenecks between the time a page was requested for from the browser till the time it is completely loaded, to determine the causes of delay.

There are several components between the browser and the Web server that could contribute to the delay — some of these can be fine tuned by you, as the architect, while for the others that are not managed by you, alternative solutions will have to be found. Let's start with a component that is not managed by you.

Network latency

Latency is caused by the underwater cables laid along the seabed and other telecom hardware. These infrastructure components contribute significantly to performance degradation. So investigations should ideally start from the

client side by measuring the latency of the Internet connection. You can do this from a terminal or command line by running *traceroute* to check potential bottlenecks. Here is what it looks like for one of our Rackspace servers hosted out of Texas in US.

With such high network latencies, it's obvious that users experience very lengthy page download periods. The quickest and probably the only way to resolve this is to ask the site visitor to access it from a faster network connection. The average latencies for servers located across the Pacific range from 300ms to 400ms. If further reduction is needed, consider moving the site from the US to a server hosted out of India, where the average latency is about 30 to 40ms. Once you are comfortable with the network latency, it's time to check for other extraneous factors affecting response speed.

Are crawlers and bots consuming too many resources?

Another rather painful behind-the-scenes cause for latency is the work of bots, crawlers, spiders and other 'digital creatures'. More often, crawlers and bots can creep into your site unnoticed, and though not all of them do so with a malicious intent, they can consume valuable resources and slow down the response times for users. In some of the corporate sites that we built, we found the Google bot had indexed the site even before